

Jiangrui Yu

Ph.D. Candidate | Efficient and Secure AI Systems
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Education

- Ph.D. Candidate in Integrated Circuit Science and Engineering** 2024-Present
School of Integrated Circuits and Institute for Artificial Intelligence, Peking University
• Advisor: Prof. Meng Li
- B.Sc. in Applied Physics** 2020-2024
School of Electronic Engineering and Computer Science, Peking University
• Dual Degree: Computer Science and Technology

Research Interests

Areas: Software-Hardware Co-Design; Privacy-Preserving Inference; LLM and Omni-Modal Inference Infrastructure; Multimodal Generation and Post-Training

Research Experience

- Ant Group**, Intern Aug 2025 – Aug 2026
Mentors: Mingzhe Zhang and Liang Kong
• Topic: Compiler and cryptographic protocol design for private AI acceleration
- Peking University**, Student Researcher – Beijing, China June 2023 – present
Institute for Artificial Intelligence and School of Integrated Circuits
• Design efficient encrypted machine learning systems through cross-layer co-design spanning cryptographic algorithms, GPU software, and specialized hardware
• Current work includes hybrid CKKS/TFHE evaluation, BatchPBS, automated HE kernel generation, and FHE acceleration

Publications

- ROSETTA: Efficient and Accurate Privacy-Preserving LLM Decoding via Hybrid CKKS/TFHE Evaluation** 2026
Jiangrui Yu, Baosheng Zhang, Liang Kong, Lin Ding, Yi Chen, Ye Yu, Mingzhe Zhang, Meng Li
ACM SIGSAC Conference on Computer and Communications Security (CCS).
- OptiPrime: Optimizing Private Inference through Protocol-Hardware Co-Design** 2026
Jiangrui Yu, Ye Yu, Si Chen, Chenqi Lin, Wenxuan Zeng, Junfeng Fan, Mingyu Gao, Meng Li
IEEE/ACM International Symposium on Microarchitecture (MICRO). **Distinguished Paper Award @ UTI 2026**
- (Invited) PEFT: A Near-Memory Processing-Enabled Heterogeneous Accelerator for BatchPBS TFHE** 2026
Jiangrui Yu, Yi Chen, Meng Li
IEEE International Symposium on Circuits and Systems (ISCAS).
- Breaking the Layer Barrier: Remodeling Private Transformer Inference with Hybrid CKKS and MPC** 2025
Tianshi Xu^{*}, Wen-jie Lu^{*}, Jiangrui Yu^{*}, Yi Chen, Chenqi Lin, Runsheng Wang, Meng Li
34th USENIX Security Symposium. ^{*} Equal contribution.
- FlexHE: A Flexible Kernel Generation Framework for Homomorphic Encryption-Based Private Inference** 2024
Jiangrui Yu, Wenxuan Zeng, Tianshi Xu, Renze Chen, Yun (Eric) Liang, Runsheng Wang, Ru Huang, Meng Li
IEEE/ACM International Conference on Computer-Aided Design (ICCAD).
- (Invited) FENIX: Flexible and Efficient Hybrid HE/MPC Acceleration with Near-Memory Processing** 2025
Tengyu Zhang, Chenqi Lin, Jiangrui Yu, Yi Chen, Shuwen Deng, Meng Li
IEEE/ACM International Conference on Computer-Aided Design (ICCAD).
- Trinity: A General Purpose FHE Accelerator** 2024
Xianglong Deng, Shengyu Fan, Zhicheng Hu, Zhuoyu Tian, Zihao Yang, Jiangrui Yu, Dingyuan Cao, Dan Meng, Rui Hou, Meng Li, Qian Lou, Mingzhe Zhang
57th IEEE/ACM International Symposium on Microarchitecture (MICRO).

Patents

- An Efficient Homomorphic Computation Method for Privacy-Preserving Inference in Neural Networks** 2026
Meng Li, Jiangrui Yu, Ye Yu, Wenxuan Zeng, Runsheng Wang
China Patent Application, CN121418079A.

Research and Competition Awards

Noteworthy Reviewer Recognition (33/222 AEC members), USENIX Security	2026
1st Place, DAC System Design Contest, DAC 2025	2025

University Honors

Doctoral Dean's Scholarship (Excellence Award), Institute for Artificial Intelligence, Peking University	2025
Outstanding Research Award, Peking University	2025
Huawei Scholarship (Third Prize)	2025
DACS Industry Contribution Award, Peking University	2024
Outstanding Graduate, Peking University and Beijing	2024
National Scholarship (Top 0.2%)	2023
Merit Student	2021-2023
Canon Scholarship	2022
Lihuirong Scholarship	2021
Bronze Medal, 35th Chinese Physics Olympiad (CPhO), as a second-year high school student	2018

Academic Service

Artifact Evaluation Committee Member (Full-term) – USENIX Security	2026
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Campus Leadership

Class Monitor – Peking University	Sept 2020 – July 2024
Deputy Director, Organizational Development Center, PKUICSS – Peking University	Sept 2022 – July 2023
Freshman Counselor – Peking University	Sept 2021